Intelligent Handover Scheme for Improved 6G NTN LEO Satellite Network Performance

Minsu Choi, Minseung Park, Junwon Kim, and Jong-Moon Chung, *Fellow, IEEE*

***Abstract*—6G non-terrestrial networks (NTNs) will use satellites for global internet access, where low earth orbit (LEO) satellites will be used frequently due to the lower latency and higher link budget that can be provided compared to higher altitude satellites. However, LEO satellites require many satellites due to their fast movement and limited coverage, leading to frequent handovers (HOs). This paper presents an intelligent LEO satellite conditional handover (ILCHO) scheme that improves the NTN performance by addressing the frequent HO issues while reducing the signal strength variation between cells and outdated measurement values. The ILCHO scheme derives the LEO satellite's state information (position and orbital path), which is used to obtain the conditions for a LEO satellite to be accessible to enable optimized target LEO satellite selection that will maximize the overall network service performance while minimizing the number of HOs. The ILCHO scheme uses multi-agent reinforcement learning (MARL) to maintain an optimized performance through early preparation by incorporating a distance-based event in the execution phase. The simulation results show that the proposed ILCHO scheme can increase the throughput and signal to noise ratio (SNR) performance compared to other HO algorithms, thereby providng improved and more stable communication conditions.**

***Index Terms*—6G, NTN, handover, LEO, MARL.**

I. Introduction

6G NETWORKS are expected to support high-performance and real-time applications (e.g., autonomous vehicles, robots, and extended reality (XR) services) as well as global coverage. As 5G wireless radio access technology (RAT) is focused on small cells, it was commonly concentrated in highly populated areas [1]. In addition, existing 3G, 4G, and 5G terrestrial networks (TNs) are vulnerable to natural disasters and unpredictable events. To overcome these issues, 6G non-terrestrial networks (NTNs) were designed to provide broadband global coverage, which includes service to less populated areas [2]. NTN consists of satellites in the low earth orbit (LEO), medium earth orbit (MEO), geostationary earth orbit (GEO), and highly elliptical orbit (HEO) as well as drone based airborne vehicle networks [3], [4], [5].

The Third Generation Partnership Project (3 GPP) Release 15 defines the objectives of NTN to provide wireless services to urban areas as well as low populated areas (e.g., remote, rural, and sub-urban areas) that are not commonly serviced by 5G networks, support reliable wireless service to highly dense Internet of Things (IoT) networks, passengers onboard moving platforms (e.g., aircraft, ships, high speed trains, etc.), and improve scalability to the network edge and user equipment (UE). The NTN specifications have been further developed in 3GPP Releases 16 and 17 [6].

LEO satellites have an altitude of 300-2,000 km, MEO satellites have an altitude above 2,000 km and below GEO satellites, where GEO satellites have an altitude of 35,786 km. Due to the altitude, the maximum propagation delay of LEO satellites is 12.89 ms, and GEO satellites is 270.73 ms. As a result, LEO satellites have lower transmission delays and higher link budgets compared to GEO satellites, but LEO satellites have much smaller coverage areas and very short coverage times due to their low altitudes compared to GEO satellites [7]. This is why many studies have investigated ultra-dense LEO satellite networks to solve the coverage issues. Fig. 1 shows the example of NTN scenario.

SpaceX began its Starlink project in 2015 and by Feburary 2025, SpaceX has launched approximately 8,000 satellites, where the number of satellites providing wireless services are about 6,100 [8], [9]. Starlink's constellation is divided into three categories: Starlink Phase 1, Starlink Phase 1 with very LEO (VLEO) satellites, and Starlink Phase 2. According to the United States Federal Communications Commission (FCC) report [10], SpaceX has submitted plans to launch a total of 4,408 satellites at altitudes between 540 and 570 km to build a network. All satellites that have been launched are operating in Phase 1. In the Starlink Phase 1 VLEO constellation [11], 7,518 satellites will operate at altitudes between 330 and 340 km. In addition, Starlink Phase 2 aims to deploy 29,998 satellites at altitudes between 340 and 614 km [12], which is currently in the planning stage. In addition, various companies such as OneWeb [13] and Amazon are working on projects for NTN. Constellation paramters used in various constellation shells are presented in Table I.

According to section 7.3.2.1 of 3GPP technical report (TR) 38.821 [7], mobility challenges due to the movement of the satellites are a big problem compared to the TN. NTN satellites move extremely fast and have long latencies compared to

Received 23 March 2025; revised 1 November 2025; accepted 1 December 2025. Date of publication 10 December 2025; date of current version 6 April 2026. This work was supported by the Korea Agency for Infrastructure Technology Advancement (KAIA) funded by the Ministry of Land, Infrastructure and Transport under Grant RS-2022-00143782 of the Republic of Korea. Recommended for acceptance by A. Abdrabou. *(Corresponding author: Jong-Moon Chung.)*

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Digital Object Identifier 10.1109/TMC.2025.3642278

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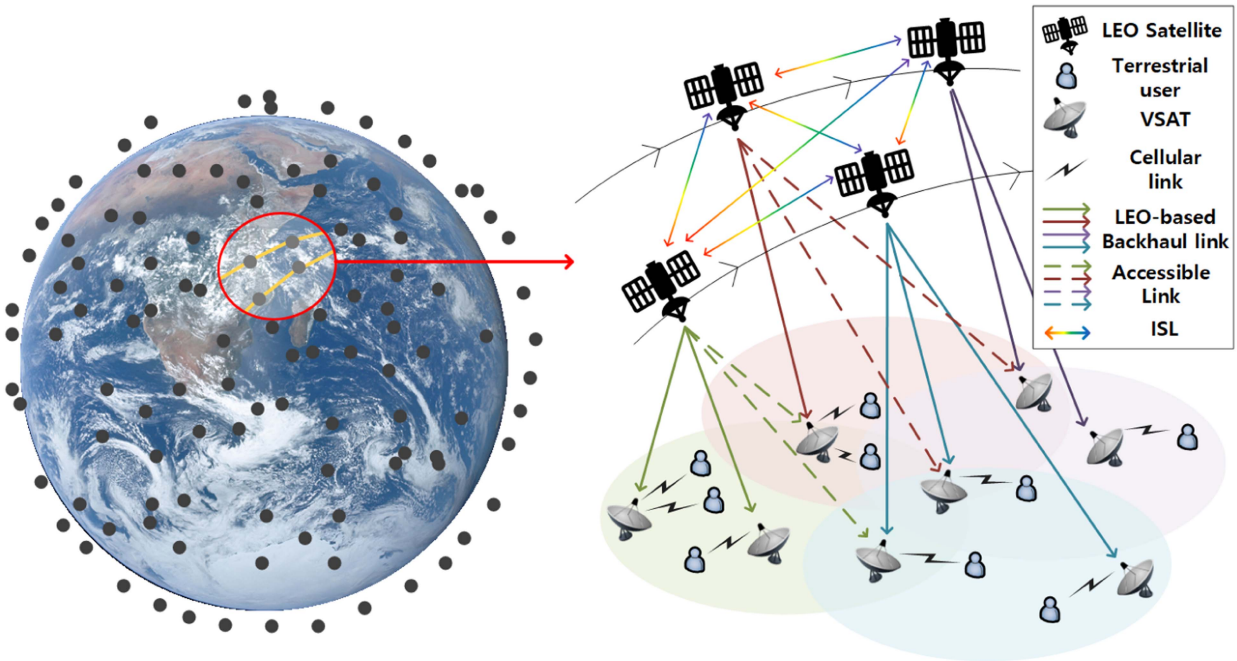

Fig. 1. Ultra dense LEO satellite NTN example.

TABLE I
LEO SATELLITE CONSTELLATION PARAMETERS USED IN THE CONSTELLATION SHELLS

Name	Altitude [km]	Inclination angle	# of orbital planes	Satellites per plane	Total satellites
Starlink Phase 1-a	550	53°	72	22	1584
Starlink Phase 2-a	340	53°	48	110	5280
OneWeb Phase 1	1200	87.9°	18	40	720

TNs, which may make the measurements (e.g., reference signals received power (RSRP), received signal strength indicator (RSSI), etc.) less reliable and UEs have difficulty in selecting the satellite to connect to due to cell overlaps and variations in the signals. In addition to these mobility-related issues, LEO satellite networks face further difficulties in ensuring reliable data transmission and efficient task execution due to intermittent connectivity and limited onboard processing capabilities [14]. Similarly, in space-air-ground integrated networks (SAGINs), the dynamic and heterogeneous topology complicates service function scheduling and adaptive resource allocation [15], while the limited energy capacity of LEO satellites introduces a critical bottleneck for continuous data transmission in emergency scenarios [16]. Furthermore, LEO satellite handover (HO) needs to be conducted every 6.61-132.28 s based on the UEs relative location in regards to the cell's diameter [7], which is why UEs experience many HO problems. As a result, 3GPP specifications state that it is not appropriate to apply conventional HO algorithms to NTNs, and proposes conditional handover (CHO) as an alternative. The triggering of a measurement report (MR) can be based on the UE's location considering four triggers, which include measurement-based triggering, location-based triggering, timing advanced value-based triggering, and elevation angle (of the source and target cells) based triggering [7].

In 3GPP Release 16, CHO has been proposed to reduce handover failure (HOF) [17]. HOF occurs mostly in the HO preparation phase, where CHO reduces HOF by making preparation decisions earlier than in baseline handover (BHO) schemes [18]. In addition, CHO was selected as a study item of NTN in 3GPP TR 38.821. Compared to these benchmarked papers, the proposed intelligent LEO satellite conditional handover (ILCHO) scheme can reduce HOF and enhance NTN service continuity, where the contributions of this paper are summarized in the following.

- *Analytic foundation of the ILCHO scheme:* The intelligent conditional handover scheme for NTNs is underpinned by three Lemmas, which serve as the algorithm's important functions. To enable accurate control through the ILCHO scheme, Lemma 1 is developed to provide state information of the LEO satellites in regards to their positions and orbital paths. Using Lemma 1, the conditions for a LEO satellite to be accessible for a given UE are specified in Lemma 2, which enables the ILCHO scheme to select the target LEO satellites for HO that will maximize the overall network service performance while minimizing the number of HOs to reduce network overhead. In addition, as NTNs need to support many LEO satellites simultaneously, Lemma 3 is provided to assist complexity and scalability estimations.

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- *Realistic constellation environment setup:* Commercially deployed or soon-to-be-deployed constellations like Starlink 1-a and Starlink Phase 2-a are used as the system model for the performance evaluation, which makes the analysis results more useful and practical.
- *Scalable MARL formulation:* Considering LEO satellite mega-constellations, the action space increases exponentially resulting in cases that the learning process does not converge and becomes unstable. For example, many papers use system models with fewer than 100 LEO satellites and set the action as the index of the reinforcement learning (RL) system, which will result in scalability issues when the number of satellites increases. To overcome this limitation, in the proposed ILCHO scheme, the action space is set to the maximum number of visible LEO satellites through Lemma 2, which reduces the action space to provide stable convergence conditions. In addition, the developed ILCHO scheme uses multi-agent reinforcement learning (MARL) to maintain an optimized performance through early preparation by incorporating a distance-based event in the execution phase.
- *RL-based validation of CHO in LEO NTNs:* To the best of the authors' knowledge, there are no existing papers that have validated RL-based CHO within a LEO satellite NTN mega-constellation. The proposed ILCHO scheme enables stable average HO counts regardless of the number of UEs being supported.

The remaining sections of this paper are organized as follows. Section II presents a literature summary of related papers. Section III introduces the LEO satellite constellation model and channel model used in the system as well as Lemmas 1, 2, and 3 followed by the formulation of the design problem. Section VI provides an overview of the ILCHO process and MARL system model. In Section V, details of the ILCHO algorithm are presented. The performance of the ILCHO scheme is analyzed in a LEO satellite NTN mega-constellation in Section VI, followed by the conclusion in Section VII.

II. Related Work

Unlike traditional networks, NTNs need to consider the high-speed movement of the LEO satellites. Therefore, related research efforts have been made to reduce signaling overhead and the unexpected power consumption caused by frequent HOs. Existing HO strategies in NTN focus on four criteria, which include, visible time, received signal strength, location, and load balance [19], [20]. The papers that achieve HO performance improvements are introduced in the following. The authors of [21] proposed a game theory-based HO strategy to maximize the UEs' benefits. The authors of [22] propose a network flow-based HO strategy to improve the quality of experience (QoE) of multiple UEs. The authors of [23] proposed a cluster-based HO persistent tunnel-based session establishment scheme, which reduces the HO procedure time.

Research efforts to apply artificial intelligence (AI) to HO control have been made in [24], [25], [26], [27], [28], where in [24] the authors use RL to improve the UE's QoE in NTNs without using a deep neural network (DNN) [29]. In [25] and [26], multi-agent Q-learning satellite HO strategies are proposed to minimize the number of HOs while not exceeding the channel budget of each satellite. However, these studies consider minimizing the number of HOs to reduce the signaling overhead, and the analysis of QoE based on the UE was not dealt in detail. Unlike other papers, [27] performs simulation based on the OneWeb constellation, where the paper focuses on throughput maximization and does not consider the signaling overhead caused by excessive HOs. The authors of [28] propose an algorithm in which during the HO preparation phase, the UE does not send an MR to the serving LEO satellite. Instead, the serving LEO satellite itself directly performs a DRL-based HO decision to select the target LEO satellite. The simulation was validated within its constellation, but the number of orbital planes was only three, and the number of actions for HO decision-making was also limited to three. When this algorithm is applied to practical constellations (e.g., Starlink or OneWeb), the effectiveness of the algorithm may decrease as its learning would have been insufficient considering the curse of dimension expansion.

However, despite these efforts, more research on performance improvement and analysis of CHO in LEO satellite NTNs is needed, as the authors of [30] point out that CHO brings mobility enhancement but can cause excessive signaling overhead due to unnecessary HOs and states that improved algorithms are needed. Especially in [31], the authors proposed an algorithm that maximizes service continuity by utilizing CHO in NTN, where the service time and capability are maximized through a service continuity performance graph model and optimal target selection algorithm.

Existing studies have conducted research on HO algorithms in the unique LEO satellite environment and have presented their performance and results. However, most of these studies utilized their constellation models, making it difficult to ensure that the performance can be highly maintained when applied to commercial mega-constellations like Starlink. Especially RL-based papers [25], [26], [27], [28] apply algorithms based on the Deep Q-network (DQN). To handle multi-user scenarios, these studies extend the single-agent into multi-agent DQN algorithms, such as the independent Q-learning (IQL) algorithm [32]. IQL enables each agent to learn separately by treating other agents as part of the environment. However, IQL may face challenges like non-stationarity due to the independent actions of multiple learning agents. Consequently, as the state space expands, convergence is not guaranteed. This issue leads to performance differences when compared with [26] through performance analysis. Furthermore, only a few studies have considered CHO in their analysis. Therefore, as far as the authors are aware, this study is the first trial to apply RL-based CHO within a LEO satellite NTN mega-constellation. The following sections describe the ultra-dense LEO satellite constellation model and demonstrate the superiority of the ILCHO scheme through an RL-based algorithm.

III. Ultra-Dense LEO SatelliteConstellation Model

In this paper, the performance of the ILCHO scheme is analyzed based on various constellation designs. The coverage

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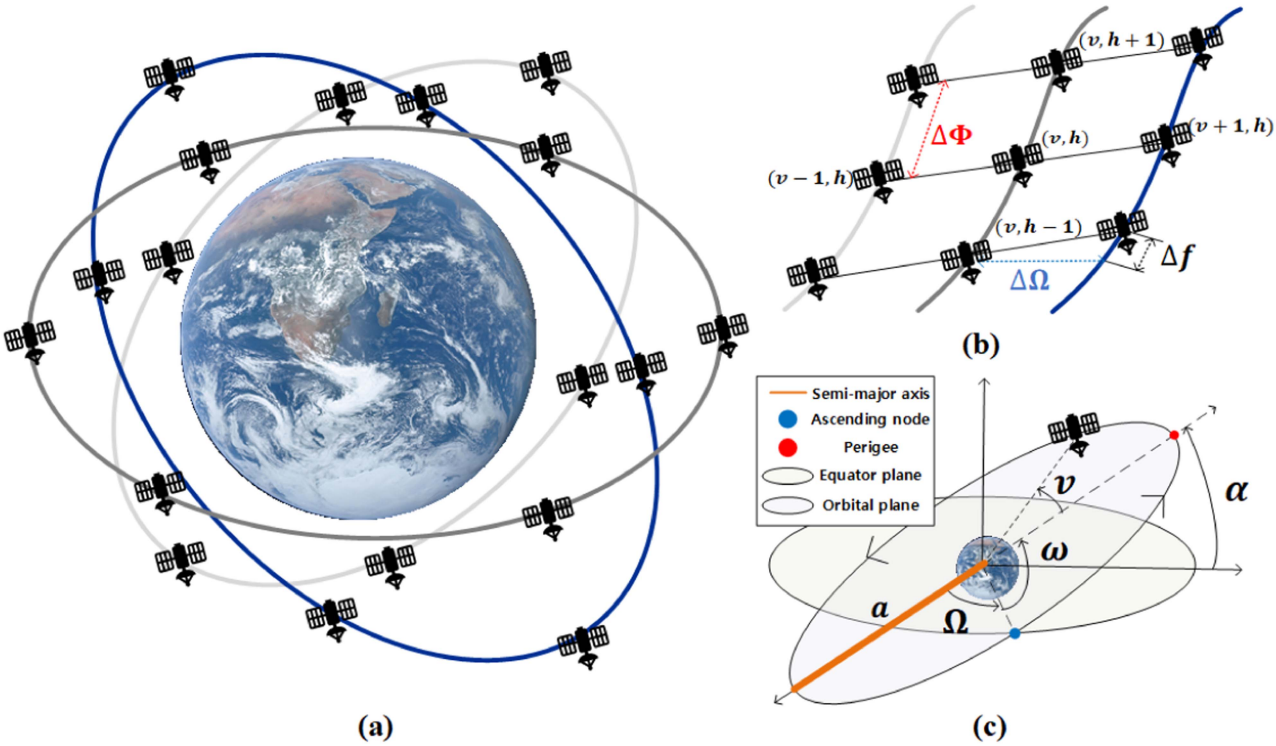

Fig. 2. LEO satellite NTN constellation topology. (a) Example of the Walker-Delta constellation $(\alpha : 18/3/F)$, (b) RAAN, phase difference between adjacent LEO satellites within a plane, (c) illustration of orbital parameters, which include the inclination angle, RAAN, argument of perigee, eccentricity, semi-major axis, and true anomaly.

of LEO satellites is determined by the minimum elevation angle ψ_{min}, altitude H_L, and constellation parameters. Due to these factors, the size of the overlapping areas between cells varies, which has a significant impact on the HO performance.

The objective of this paper is to maximize the NTN service performance through the proposed ILCHO scheme. In the following, first, a description of the constellation model for LEO satellites is provided in Section II-A, followed by the mathematical definition of the network and channel model in Section II-B, which is followed by the optimization problem setup in Section II-C.

In the model, there are M LEO satellites and K UEs, where the set of LEO satellites $S = \{s_m \mid m \in [1, M]\}$ are equipped with a regenerative payload that can support base station functions with on-board hardware. The set of UEs is denoted as $U = \{u_k \mid k \in [1, K]\}$ and L is defined as the set of the service indications between the UEs and LEO satellites, which is expressed as $L = \{l_{m,k} \mid m \in [1, M], k \in [1, K]\}$, where $l_{m,k} \in 0, 1$ in which $l_{m,k} = 1$ means that UE u_k is being serviced by the mth LEO satellite.

A. Constellation Model

As illustrated in Fig. 2, in this paper, the constellation model is based on a Walker-Delta model, where all orbits have the same inclination angle α. The Walker-Delta constellation has N_p orbital planes and N_L LEO satellites per orbital plane, where each orbital plane is separated by the same angular distance of $2\pi/N_p$. The total number of the LEO satellites is $N_{total} = N_p N_L$. The right ascension of the ascending node (RAAN) is $\Delta\Omega = 2\pi/N_p$ where each orbital plane is uniformly distributed in the equatorial plane. N_L LEO satellites are evenly distributed within one plane and the phase difference between two adjacent LEO satellites is expressed as $\Delta\Phi = 2\pi/N_L$. In addition, the phase difference between LEO satellites in adjacent planes is calculated by $\Delta f = 2\pi F/N_{total}$ where F is the phasing factor [33]. Therefore, the Walker-Delta constellation is represented by $\alpha : N_{total}/N_p/F$. In addition, as shown in Fig. 2(b), the mth LEO satellite corresponds to the hth LEO satellite of the vth orbital plane. Therefore, m is represented by a logical index (v, h), where $m = (v-1)N_L + h$, in which $1 \le v \le N_p$ and $1 \le h \le N_L$. The orbit of a LEO satellite is operated at an altitude of H_L, where there are seven values of ephemeris data used to determine information about the orbital planes and the exact location of the LEO satellites, which include the inclination angle α, RAAN, eccentricity ϵ, semi-major axis a_{s}, argument of perigee w_{perigee}, mean anomaly A, and ephemeris reference time t_{0e} [7].

The semi-major axis $a_{\text{s}} = R_E + H_L$ is half of the major-axis diameter in an elliptical orbit, where R_E is the earth radius. The argument of perigee w_{perigee} is the angle between the ascending node and the perigee. The ascending node is the reference point where the LEO satellite crosses the reference plane in an upward direction. It is the intersection of the reference plane and the orbital plane where the satellite is moving upwards.

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The position of the LEO satellites follows an eliptical orbital path. The positions of the LEO satellites can be obtained based on the ephemeris data and constellation parameters defined above. According to 3GPP 38.300 [34], semi-major axis, argument of perigee, inclination angle, eccentricity, RAAN, and phasing factor are determined by the constellation models like Starlink Phase 1-a and Phase 2-a. In addition, the true anomaly can be calculated with phasing parameters and t_{0e} is provided in the system information block (SIB19). In this study, the LEO satellites' position is considered over a duration of T time slots, denoted by $t = 0, 1, \ldots, T-1$. In time slot t, LEO satellite m's position vector can be represented as $s_m^t = (x_m^t, y_m^t, z_m^t)$, and the angular parameter true anomaly is $\mu^t = \mu^0 + \omega t$, where the LEO satellite's angular speed is $\omega = \sqrt{(GM_E/a_s^2)}$, in which G is the gravitational constant, and M_E is the earth's mass. An object moving along a Kepler orbit, rather than a circular one, tracks its position based on the value of the true anomaly. The state of the mth LEO satellite (at time step t) s_m^t can be obtained through Lemma 1.

Lemma 1: The cartesian coordinate that represent the state of LEO satellite m (at time t) s_m^t is represented as

$$s_m^t = a_s \begin{pmatrix} \cos\psi_m^t \cos\Omega_m - \sin\psi_m^t \sin\Omega_m \cos\alpha \\ \cos\psi_m^t \sin\Omega_m + \sin\psi_m^t \cos\Omega_m \cos\alpha \\ \sin\psi_m^t \sin\alpha \end{pmatrix} \tag{1}$$

where $\psi_m^t = \Phi_m + \omega t + f_0$. □

Proof: The coordinates of N_L LEO satellites uniformly distributed within an orbital plane with an inclination angle of $0°$ can be expressed as

$$P_j = \begin{bmatrix} x_j \\ y_j \\ z_j \end{bmatrix} = \begin{bmatrix} r_j \cos\mu_j \\ r_j \sin\mu_j \\ 0 \end{bmatrix}$$

where μ_j and r_j are the true anomaly and semi major axis of the jth LEO satellite, repectively. By rotating the orbit P_j by the inclination angle α and then rotating it by the RAAN $\Omega_v (= \Omega_m)$, the vth LEO satellite in the hth orbital plane at the initial state can be obtained and is expressed as

$$s_m^0 = s_{(v,h)}^0 = R_z(\Omega_v) R_x(\alpha) \begin{bmatrix} r_h \cos\mu_h \\ r_h \sin\mu_h \\ 0 \end{bmatrix}$$

where $R_x(\alpha)$ is the rotation matrix that rotates the orbital plane by the inclination angle α about the x-axis, and $R_z(\Omega_v)$ is the rotation matrix that rotates the orbital plane by the RAAN $\Omega_v = \Delta\Omega(v-1)$ of the vth orbital plane about the z-axis. These matrices are calculated as follows.

$$\begin{aligned} s_m^0 = s_{(v,h)}^0 &= \begin{bmatrix} \cos\Omega_v & -\sin\Omega_v & 0 \\ \sin\Omega_v & \cos\Omega_v & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\alpha & -\sin\alpha \\ 0 & \sin\alpha & \cos\alpha \end{bmatrix} \\ &\quad \times \begin{bmatrix} r_h \cos\mu_h \\ r_h \sin\mu_h \\ 0 \end{bmatrix} \\ &= r_h \cos\mu_h \cos\Omega_v - r_h \sin\mu_h \sin\Omega_v \cos\alpha \\ &\quad + r_h \cos\mu_h \sin\Omega_v + r_h \sin\mu_h \cos\Omega_v \cos\alpha \\ &\quad + r_h \sin\mu_h \sin\alpha \end{aligned}$$

The true anomaly at time step t, μ_m^t (or μ_h^t for $s_{(v,h)}^t$) is calculated using the mean anomaly A_m^t and the eccentric anomaly E_m^t, and the relationship is given as

$$A_m^t = \Phi_h + \omega t$$

$$A_m^t = E_m^t - e \sin E_m^t$$

$$\cos\mu_m^t = \frac{\cos E_m^t - \epsilon}{1 - \epsilon \cos E_m^t}$$

where $\Phi_h = \Delta\Phi h = 2\pi h / N_L$. The distance between the mth LEO satellite and the Earth's center r_m^t is given as follows.

$$r_m^t = \frac{a_s(1-\epsilon^2)}{1 + \epsilon \cos\mu_m^t}$$

In this paper, simulation has been conducted within the Starlink constellation environment. The eccentricity of the Starlink constellation is 0.0001379. Therefore, eccentricity can be approximated to be zero. The true anomaly and the mean anomaly are considered to be identical for $\epsilon = 0$. Therefore, by adding the initial phase parameter f_0 and replacing the logical index with m, the coordinates of the mth LEO satellite at time step t are derived as follows.

$$\begin{aligned} s_m^t &= \frac{a_s\left(1-\epsilon^2\right)}{1 + \epsilon \cos v_m^t} \begin{bmatrix} \cos\mu_m^t \cos\Omega_v - \sin\mu_m^t \sin\Omega_v \cos\alpha \\ \cos\mu_m^t \sin\Omega_v + \sin\mu_m^t \cos\Omega_v \cos\alpha \\ \sin\mu_m^t \sin\alpha \end{bmatrix} \\ &= a_s \begin{bmatrix} \cos\left(\Phi_m + \omega t + f_0\right) \cos\Omega_m \\ -\sin\left(\Phi_m + \omega t + f_0\right) \sin\Omega_m \cos\alpha \\ \cos\left(\Phi_m + \omega t + f_0\right) \sin\Omega_m \\ +\sin\left(\Phi_m + \omega t + f_0\right) \cos\Omega_m \cos\alpha \\ \sin\left(\Phi_m + \omega t + f_0\right) \sin\alpha \end{bmatrix} \end{aligned}$$

■

The latitude and longitude of UE k are respectively denoted by ϕ_k and δ_k. The UE's position state $u_k = (x_k, y_k, z_k)$ is expressed considering the UE's antenna configuration, which is assumed to be a very small aperture terminal (VSAT) that does not change its position over time. Due to the significantly higher speed of LEO satellites (7.59 km/s at an altitude of 550 km) compared to the UEs, the UE's mobility is often considered negligible [35], [36]. Therefore, fixed UE models are commonly used in prior studies, and the same assumption is adopted in this work [19], [20], [22], [25], [26], [27], [28], [30], [31], [37].

$$\mathbf{u}_k = \begin{pmatrix} x_k \\ y_k \\ z_k \end{pmatrix} = R_E \begin{pmatrix} \cos\phi_k \cos\delta_k \\ \cos\phi_k \sin\delta_k \\ \sin\phi_k \end{pmatrix} \tag{2}$$

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The distance between UE k and LEO satellite m is calculated from (3) [38].

$$\begin{aligned} d_{m,k} &= \|\mathbf{s}_m - \mathbf{u}_k\| \\ &= \sqrt{R_E^2 \sin^2 \psi_{m,k} + H_L^2 + 2H_L R_E} - R_E \sin \psi_{m,k} \end{aligned} \tag{3}$$

The position of the LEO satellite is based on a predefined deterministic motion. Given the assumption that the UE has prior knowledge of the ephemeris data, $d_{m,k}$ can be calculated based on the elevation angle $\psi_{m,k}$ between UE k and LEO satellite m. The elevation angle between UE k and LEO satellite m can be calculated as follows.

$$\psi_{m,k} = \frac{\pi}{2} - \arccos\left(\frac{\mathbf{u}_k \cdot (\mathbf{s}_m - \mathbf{u}_k)}{\|\mathbf{u}_k\|\|\mathbf{s}_m - \mathbf{u}_k\|}\right) \tag{4}$$

In this paper, if a UE is within the coverage of any LEO satellite, it is referred to as being in an 'accessible' state. Therefore, the condition for LEO satellite m to be accessible for UE k is specified in Lemma 2.

Lemma 2: When the condition of (5) is satisfied, UE u_k is accessible LEO satellite s_m

$$(c_1 \cos \mu_m + c_2 \sin \mu_m) \geq \frac{d_{m,k} \sin \psi_{\min}}{R_E + H_L} \tag{5}$$

where $c_1 = \cos \phi_k \cos(\Omega - \delta_k)$ and $c_2 = \sin \alpha \sin \phi_k - \cos \alpha \cos \phi_k \sin(\Omega - \delta_k)$. □

Proof: The condition for LEO satellite s_m to be accessible for UE u_k is that $\psi_{m,k} \geq \psi_{\min}$. Substituting $\psi_{m,k}$ into the previous equation as defined in (4) yields the following result.

$$\frac{\mathbf{u}_k \cdot (\mathbf{s}_m - \mathbf{u}_k)}{\|\mathbf{u}_k\|\|\mathbf{s}_m - \mathbf{u}_k\|} \geq \sin \psi_{\min} \tag{6}$$

The numerator is calculated as follows.

$$\mathbf{u}_k \cdot (\mathbf{s}_m - \mathbf{u}_k) = \mathbf{u}_k \cdot \mathbf{s}_m - R_E^2$$

The inner product of $\mathbf{u}_k$ and $\mathbf{s}_m^t$ is calculated as follows.

$$\begin{aligned} \mathbf{u}_k \cdot \mathbf{s}_m = R_E(R_E + H_L) \{&(\cos \Omega_m \cos \mu_m \\ &- \sin \Omega_m \sin \mu_m \cos \alpha) \\ &\times \cos \phi_k \cos \delta_k \\ &+ (\sin \Omega_m \cos \mu_m + \cos \Omega_m \sin \mu_m \cos \alpha) \\ &\times (\cos \phi_k \sin \delta_k) \\ &+ (\sin \mu_m \sin \alpha)(\sin \phi_k)\} \\ = \{&\cos \mu_m \cos \phi_k (\cos \Omega \cos \delta_k - \sin \Omega \sin \delta_k) \\ &- \sin \mu_m \cos \alpha \cos \phi_k (\sin \Omega \cos \delta_k - \cos \Omega \sin \delta_k) \\ &+ \sin \mu_m \sin \alpha \sin \phi_k\} \\ = \{&\cos \mu_m \cos \phi_k \cos (\Omega_m - \delta_k) \\ &- \sin \mu_m \cos \alpha \cos \phi_k \sin (\Omega_m - \delta_k) \\ &+ \sin \mu_m \sin \alpha \sin \phi_k\} \\ = (&c_1 \cos \mu_m + c_2 \sin \mu_m) \end{aligned}$$

Substituting (6) with the previously obtained results, the inequality of (7) can be obtained through rearrangements.

$$\frac{R_E(R_E + H_L)(c_1 \cos \mu_m + c_2 \sin \mu_m)}{\|\mathbf{u}_k\| d_{m,k}} \geq \sin \psi_{\min} \tag{7}$$

The phasing factor μ_m is a time-varying variable determined by angular parameters. At each time stamp, it can be verified if LEO satellite s_m is accessible for UE u_k using (5). ■

According to Lemma 2, the deterministic movement of LEO satellites allows the set of accessible satellites for user u_k to be calculated using a phasing parameter in advance. This enables UE u_k to accurately estimate the number of accessible LEO satellites based on their latitude. Pre-calculating accessibility with ephemeris data from LEO satellites in advance significantly reduces the computational cost by eliminating the need to identify all users and LEO relationships. In Section III-B, the LEO channel model is described in detail. UE u_k can select the optimal LEO satellite among the accessible LEO satellites using the ILCHO scheme proposed in this paper.

B. LEO Network and Channel Model

Based on 3GPP TR 38.811 and TR 38.821, the path loss between two nodes can be modeled as

$$P_{\text{loss}} = P_{\text{loss},b} + P_{\text{loss},g} + P_{\text{loss},s} + P_{\text{loss},e} \tag{8}$$

where $P_{\text{loss},b}$ is the basic path loss and $P_{\text{loss},g}$ is the attenuation due to atmospheric gases and rain fades. $P_{\text{loss},s}$ is the scintillation loss due to the ionospheric or tropospheric characteristics of the environment, in which $P_{\text{loss},s} = 0$ for latitudes between $\pm 20°$ and $\pm 60°$. $P_{\text{loss},e}$ represents the building entry loss in dB. In the LEO satellite model, considering that the UE is located outdoors, the path loss $P_{\text{loss},e}$ can be neglected if the link between a UE and LEO satellite has line of sight (LoS). $P_{\text{loss},b}$ is modeled as

$$P_{\text{loss},b} = F_{\text{loss}}(d, f_c) + S_F + C_L(\alpha_{C_L}, f_c) \tag{9}$$

where $F_{\text{loss}}(d, f_c)$ is the free space path loss, $C_L(\alpha_{C_L}, f_c)$ is the clutter loss, S_F is shadow fading loss, which is modeled from a normal distribution generated random number $S_F \sim N(0, \sigma_{S_F}^2)$, and $F_{\text{loss}}(d, f_c)$ is modeled as (10).

$$F_{\text{loss}}(d, f_c) = 32.45 + 20 \log_{10}(f_c) + 20 \log_{10}(d) \tag{10}$$

Shadow fading and clutter loss depend on the scenario and the LoS conditions. The values of C_L and S_F are used from Table 6.6.2 of TR 38.811 [6]. Atmospheric attenuation is modeled as follows $P_{\text{loss},g} = L_{\text{zenith}}(f_c)/\sin \psi$, where $L_{\text{zenith}}(f)$ is the zenith attenuation for frequencies between 1 and 350 GHz and ψ is the elevation angle. These attenuation factors collectively affect the overall link quality and mobility performance in the LEO environment. Specifically, path loss primarily governs the received power and consequently determines the duration of visible links and the frequency of handover events. Shadow fading introduces large-scale fluctuations that increase the likelihood of handover failures, while small-scale fading causes rapid power variations that degrade the throughput under high mobility conditions.

In this paper, Doppler shift is not considered. While Doppler shift can potentially impact LEO satellite communications,

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3GPP specifies [6] that its effects can be effectively compensated and are not regarded as a major impairment in practical NTN scenarios. As noted in [6], most Doppler shift and its variation can be compensated by either pre- or post-processing, based on known satellite trajectories and UE positions. The NR physical layer is also designed to operate reliably under such conditions by adapting parameters such as subcarrier spacing and DMRS configuration. Moreover, numerous studies [39], [40], [41] have demonstrated that Doppler effects in LEO satellite communications can be effectively mitigated through both hardware and software-based techniques.

The overall path loss is composed of the weighted sum of the LoS and non-LoS (NLoS) channels. Therefore, the path loss between UE k and LEO satellite m (i.e., $P_{\text{loss},m,k}$) is

$$P_{\text{loss},m,k} = P_{\text{LoS}} P_{\text{loss}} + (1 - P_{\text{LoS}}) P_{\text{loss, NLoS}} \tag{11}$$

where P_{LoS} is the probability of LoS, P_{loss} is the path loss of the LoS signal, and $P_{\text{loss, NLoS}}$ is the path loss of the NLoS signal. Using (11), the signal-to-noise ratio (SNR) between UE k and LEO satellite m can be represented as

$$\gamma_{m,k} = E_{EIRP} + G_T - K_B - P_{\text{loss},m,k} - B_W \tag{12}$$

where E_{EIRP} is the equivalent isotropic radiated power, K_B is the Boltzmann constant, G_T is the received antenna gain-to-noise temperature ratio, and B_W is the bandwidth [7].

According to the Shannon capacity theorem, the throughput volume that LEO satellite m provides to UE k can be obtained from (13).

$$R_{m,k} = \begin{cases} B_{W,m} \log_2(1 + \gamma_{m,k}), & l_{m,k} = 1 \\ 0, & l_{m,k} = 0 \end{cases} \tag{13}$$

In this paper, the ILCHO scheme's objective is to maximize the overall network service performance. Therefore, the objective function for the NTN service performance is defined as R_{NTN}, where R_{NTN} is defined as the total average sum of the throughput received by all UEs over the total time duration T, which is expressed as follows.

$$R_{\text{NTN}} = \frac{\sum_{t \in T} \sum_{s_m \in S} \sum_{u_k \in U} R_{m,k}}{K} \tag{14}$$

C. Problem Formulation

The goal of the ILCHO scheme is to solve the optimization problem to maximize the NTN service performance, given that multiple UEs effectively share the limited resources that are accessible from the LEO satellites. Therefore, the optimization statement to maximize the NTN performance is formulated as

$$\begin{aligned} (\mathcal{P}) \quad & \max \quad R_{NTN} \\ & \text{subject to} \quad (5) \\ & \sum_{m=1}^{M} l_{m,k} \le 1, \quad \forall k \qquad (15a) \\ & \sum_{k=1}^{K} l_{m,k} \le J, \quad \forall m \qquad (15b) \end{aligned}$$

where (5) is from Lemma 2, which is the coverage constraint that LEO satellite m is considered as an accessible LEO satellite for UE k based on Lemma 1. In addition, (15a) indicates that UE u_k can communicate with at most one LEO at a time, and (15b) states that LEO satellite s_m^t can communicate with up to J UEs, because its bandwidth can be divided into J channels. The complexity of the optimization problem $(\mathcal{P})$ is analyzed in the following Lemma 3.

Lemma 3: For a fixed $t = t_0$ and given accessibility sets derived from Lemma 2, the optimization problem $(\mathcal{P})$ is equivalent to a generalized assignment problem and is therefore NP-hard. □

Proof: Let the $(\mathcal{P}')$ be the static version of $(\mathcal{P})$ for a fixed $t = t_0$. At the time step t_0, the set of accessible LEO satellites $\mathcal{S}_m$ for each UE u_k and the potential throughput $R_{m,k}$ for each possible link are considered fixed. The problem is to find a binary assignment $l_{m,k}$ that maximizes the total instantaneous throughput. The static problem $(\mathcal{P}')$ is formulated as follows.

$$\begin{aligned} max \quad & \sum_{m=1}^{M} \sum_{k=1}^{K} R_{m,k} \\ & \sum_{m=1}^{M} l_{m,k} \le 1, \quad \forall k \\ & \sum_{k=1}^{K} l_{m,k} \le J, \quad \forall m \qquad (16a) \\ & l_{m,k} \in 0, 1 \qquad (16b) \end{aligned}$$

This problem $(\mathcal{P}')$ is a direct instance of the classic NP-hard generalized assignment problem. The full optimization problem $(\mathcal{P})$ requires finding an optimal policy over a sequence of these NP-hard static assignment problems across the time duration T. Because the single-instant problem $(\mathcal{P}')$ is already NP-hard, time-varying problem $(\mathcal{P})$ is also NP-hard. ■

IV. Preliminaries

A. Overview of the ILCHO Process

3GPP proposed a new HO process called CHO in Rel-16 TS 38.300 to improve mobility in 5G TNs [34]. In the 3GPP BHO procedures, the network oversaw the decision-making process of when to conduct HO, which is a reactive process that tends to cause HOF. On the other hand, CHO allows the UE to decide when to perform HO when certain conditions are met, thereby enabling more proactiveness. There are two independent events for CHO, which are the 'preparation event' and 'execution event' which are described in the following.

Preparation phase: If the preparation event is satisfied, the UE sends a MR to the source LEO (S-LEO) satellite. In the ILCHO process, the optimal target LEO (T-LEO) satellite is selected using the ephemeris data of the LEO satellites obtained through Lemmas 1 and 2 and the state information of the LEO satellites is calculated based on the UE location information. This decision is made through the MARL control algorithm, in which details are provided in Algorithm 1. The S-LEO satellite

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configures the candidate T-LEO satellites and sends a CHO request message to each of the T-LEO satellites, and will receive an acknowledgment (ACK) in return. The S-LEO satellite then sends a CHO command to the UE, where there is a hold-off time before the execution phase, during which the T-LEO satellites reserve radio resources.

Execution Phase: When the T-LEO satellite satisfies the execution event, the UE sends a HO inform message to the S-LEO satellite. The S-LEO satellite forwards the downlink (DL) data to the T-LEO satellite. The T-LEO satellite performs synchronization with the UE, followed by uplink (UL) allocation and random access (RA). The UE then sends a CHO confirm message to the T-LEO satellite. In the ILCHO process, the distance-based event in (28) is used.

Completion Phase: This is the same as the BHO procedure. In the core network, the bearer is modified, and the path is switched. The T-LEO satellite sends a HO complete message to the S-LEO satellite and then the CHO process is completed. In this study, a DRL-based algorithm is used during the preparation phase to appropriately select the T-LEO satellite, and a distance-based event A3 is utilized during the execution phase to perform CHO evaluation. The proposed ILCHO scheme extends the traditional CHO scheme, where MARL is applied as a controller, as presented in the following.

B. Backgrounds of Reinforcement Learning

The ILCHO scheme uses a MARL system that is based on a Markov Decision Process (MDP) developed to assist the HO control process. The MDP is composed of a state s_t and an action a_t. The agent observes the current state s_t (or s) in the environment and then takes an action a_t (or a) according to policy π. Then, the agent will obtain a reward r_t (or r) and the next state s_{t+1} (or s') is produced. Through the process of interacting with the environment, the goal of the RL is to find an optimal policy that maximizes the accumulative reward $R_t = \sum_{i=t}^{T_{\text{final}}} \sigma^{i-t} r_i$ where $\sigma \in (0, 1]$ is the discount factor and T_{final} is the final time slot. $Q(s_t, a_t)$ (or $Q(s, a)$) is the action-value function which is the expected return obtained by taking action a_t at the current state s_t, that is represented as $Q(s, a) = \mathbb{E}[R \mid s, a]$. According to the Bellman optimality equation, the optimal action-value function $Q^*(s, a) = \max_{a^*} Q(s, a)$ is calculated as follows.

$$Q^*(s, a) = \mathbb{E}\left[r + \sigma \max_{a'} Q^*(s', a') \,|\, s, a\right] \tag{17}$$

Q-learning is used to find the optimal value by iteratively updating the Q-value through temporal difference (TD) learning (where β is the learning rate) based on (17).

$$Q(s, a) \leftarrow Q(s, a) + \beta \left[r + \sigma \max_{a'} Q^*(s', a') - Q(s, a)\right] \tag{18}$$

The DQN algorithm uses a DNN in the RL process with the action-value function $Q(\cdot)$ [42]. In DQN, the $Q(\cdot)$ value is obtained through the DNN output and the network parameter θ^{DQN} is trained with a loss function $L^{\text{DQN}}(\theta^{\text{DQN}}) = \mathbb{E}[(y - Q(s, a \mid \theta^{\text{DQN}}))^2]$, where $y = r + \sigma \max_{a'} Q(s', a' \mid \theta^{\text{DQN}})$, which is generated by the DNN, given the state s'.

In addition, DQN uses experience replay and a target network to enhance the stability of learning. Experience replay generates a buffer $\mathcal{D}$ and stores the tuple (s, a, r, s') for each transition. The experience replay relies on mini batches, using randomly selected samples for learning, which helps to reduce the correlation between consecutive transitions. In addition, a target network with the same structure as the main Q-network is created. The target network's weights are updated at certain intervals. The update method is based on

$$\theta^{\text{DQN}'} \leftarrow \eta_{\text{DQN}} \theta^{\text{DQN}'} + (1 - \eta_{\text{DQN}}) \theta^{\text{DQN}'} \tag{19}$$

where θ^{DQN} and $\theta^{\text{DQN}'}$ are respectively the parameters of the main Q-network and target network, and η_{DQN} is the soft update coefficient.

C. QMIX Algorithm for MARL

In this paper a value-based MARL algorithm QMIX is used [32]. Among various MARL algorithms, MADDPG was not adopted due to its relatively limited performance in cooperative scenarios and scalability challenges. Although MAPPO demonstrates a strong performance, the QMIX algorithm was chosen in this paper as it exhibits superior learning efficiency and faster convergence [43]. The system model involves multiple UEs acting as agents, which is modeled as a multi-agent extension of MDPs.

In this work, the role of an agent is explicitly assigned to each UE. Operating with incomplete information, each UE cannot fully observe the other UEs. The QMIX algorithm operates in centralized training and decentralized execution (CTDE) mode. During training, a central controller utilizes the global information of all agents for learning. This centralized training allows for joint action optimization across agents. Through the monotonic mixing network in QMIX, global consistency is maintained even when each agent selects its own locally optimal action. In the execution phase, agent u_k performs actions based on its observation o_k, which is limited to its observation range. The execution phase is inherently a decentralized partially observable Markov decision process, where the CTDE paradigm trains on the joint observations and actions of all agents, while QMIX's monotonic value factorisation ensures that the learned policy remains optimal under local observations. In this paper, it is assumed that the centralized training is carried out by the ground based near real-time (RT) RAN intelligence controller (RIC) [44]. The near RT-RIC aggregates these observations and the ephemeris data of the LEO satellites, and carries out centralized training. In addition, the onboard system executes the inference process (CHO commands).

The validity of satellite ephemeris data is governed by the T430 timer. As specified in 3GPP TS 38.331 [45], this timer is configured by the ntn-UlSyncValidityDuration parameter, which sets the validity period of the satellite's orbital information. This duration is not measured in milliseconds, but on a much larger scale. As specified in 3GPP TS 38.331, the ntn-UlSyncValidityDuration parameter is enumerated with the possible values of s5, s10, s15, s20, s30, s40, s50, s60, s90, s120, s180, s240, s300, min10, min15, min20 which range from 5 seconds to 20 minutes. Therefore, the signaling delay is ignored

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in this paper because the one-way latency for a LEO satellite at an altitude of 550 km is approximately 5 ms.

Agent k's action-value function is denoted as $Q_k(\tau_k, a_k)$, where a_k is the joint action of agent k and $\tau_k = (o_k^1, \ldots, o_k^t)$ is the local observation history of agent k. The joint action-value function $Q_{\text{tot}}(\boldsymbol{\tau}, \boldsymbol{a})$ represents the overall system's joint action value, where $\boldsymbol{\tau} = (\tau_1, \ldots, \tau_K)$ and $\boldsymbol{a} = (a_1, \ldots, a_K)$ are respectively the joint observation history and joint actions of all agents. The result of the argmax operation on $Q_{\text{tot}}(\boldsymbol{\tau}, \boldsymbol{a})$ is made to align with the result of the argmax operation performed on an individual agent $Q_k(\cdot)$. This ensures that during the execution phase, agents can select greedy actions based on $Q_k(\cdot)$ using only the information available within their local observation range.

$$\arg\max_{a} Q_{\text{tot}}(\boldsymbol{\tau}, \boldsymbol{a}) = \begin{bmatrix} \arg\max_{a_1} Q_1(\tau_1, a_1) \\ \vdots \\ \arg\max_{a_K} Q_K(\tau_K, a_K) \end{bmatrix} \tag{20}$$

However, if Q_{tot} is represented as a linear sum of individual value functions Q_k (i.e, $Q_{tot}(\boldsymbol{\tau}, \boldsymbol{a}) = \sum_{k=1}^{K} Q_k(\tau_k, a_k)$) as in [46], the training procedure cannot fully exploit the other agents' observations. To address this limitation, QMIX preserves the 'global $\arg\max$ = independent $\arg\max$' property but replaces the simple linear sum with a monotonic mixing network, redefining Q_{tot} is derived as follows.

$$\arg\max_{a} Q_{\text{tot}}(\boldsymbol{\tau}, \boldsymbol{a}) = f_{\text{mix}}(Q_1, \ldots, Q_K) \tag{21}$$

$$\frac{\partial Q_{\text{tot}}}{\partial Q_k} \geq 0, \quad \forall k \tag{22}$$

By representing Q_{tot} with the function of Q_k with the following monotonicity constraint between Q_{tot} and Q_k, Q_{tot} can be obtained. This constraint (22) ensures that the global action-value function Q_{tot} is a monotonic function of each agent's action-value function Q_k. This property is crucial for mitigating the non-stationarity problem, as it guarantees that a greedy action selected by an agent locally will also lead to an increase in the global Q_{tot}. This provides a consistent and stable learning signal for each agent, even as other agents' policies evolve.

The QMIX algorithm consists of three neural networks, which are the agent network, which represents Q_k, the mixing network, which combines Q_k into Q_{tot}, and the hypernetwork. Agent u_k has an agent network, where the current local observation o_k^t and the previous action a_k^{t-1} are the inputs to the agent network, which produces the individual Q-value $Q_k(\tau_k^t, a_k^t)$ for the agent.

The agent network adopts a deep recurrent Q-network (DRQN) structure. The mixing network takes the output of the agent network $Q_k(\tau_k^t, a_k^t)$ as input. The outputs of the mixing network are used in the joint action-value function $Q_{\text{tot}}(\boldsymbol{\tau}, \boldsymbol{a})$. Inside the mixing network, the hypernetwork generates the weights for the mixing network to ensure that the monotonicity constraint is satisfied. The input to the hypernetwork is the local observation of all agents, and the output is the weights for the mixing network. The overall structure is illustrated in Fig. 4.

Algorithm 1: ILCHO Algorithm.

Initialize: evaluation network θ_{QMIX}, target network θ_{QMIX}^{-}, mixing network, hyper-networks;
Inputs: M, K, constellation parameters $(N_{\text{total}}, N_p, F, \alpha, H_L, t_{0e}, \mu)$, $N_{\max}, \psi_{\min}, L$, and hyper-parameters $(\beta, \sigma, \epsilon, b, N_{\text{epi}}, T_{\text{t}}, \eta_{\text{QMIX}})$
for $n = 1$ to N_{epi} **do**
  Initialize LEO satellite environment (u_k, s_m)
  for $t = 1$ to T_{serv} **do**
    Obtain the observation of each agent k: o_k^t
    for each agent $k = 1$ to K **do**
      Obtain the observation o_k^t
      Choose action a_k^t using the ϵ-greedy policy
      Choose optimal T-LEO satellite selection in the Preparation phase
      if selected LEO satellites satisfy (28) **then**
      (i.e., Execution events are met)
      Execute HO
        Obtain the reward r_k^t and next observation o_k^{t+1}
        Store $(o^t, \tau^t, a^t, r^t, o^{t+1}, \tau^{t+1})$ in replay buffer $\mathcal{D}$
        end if
      end for
    Sample minibatches b from $\mathcal{D}$ replay buffer
    Obtain $\hat{a}$ through (21)
    Compute target value (24) and the loss function (23)
    Update the network by minimizing the loss (23)
    if $(t \mod T_{\text{t}}) == 0$ **then**
      Update the target network (25)
    end if
  end for
end for

The QMIX algorithm aims to maximize the joint Q-value Q_{tot} to achieve a shared objective. The learning process involves minimizing the QMIX loss function $L_{\text{QMIX}}(\theta_{\text{QMIX}})$, where θ_{QMIX} represents the parameters of the model. The loss function $L_{\text{QMIX}}(\theta_{\text{QMIX}})$ and the TD target y_{QMIX} are defined as

$$L_{\text{QMIX}}(\theta_{\text{QMIX}}) = \sum_{b} \left[(y_{\text{QMIX}} - Q_{\text{tot}}(\boldsymbol{\tau}, \boldsymbol{a}, s; \theta_{\text{QMIX}}))^2 \right] \tag{23}$$

$$y_{\text{QMIX}} = r + \sigma \max_{a'} Q_{\text{tot}}(\boldsymbol{\tau}^{-}, \boldsymbol{a}^{-}, s^{-}; \theta_{\text{QMIX}}^{-}) \tag{24}$$

where b is the batch size of the replay buffer, r is the reward of action, and σ is the discount factor. The parameters $\boldsymbol{\tau}^{-}, \boldsymbol{a}^{-}, s^{-}$, and θ_{QMIX}^{-} represent the joint action observation history, joint action, state of QMIX, and the parameters of the target network in the QMIX, respectively. The target network is updated every T_t interval through (25).

$$\theta_{\text{QMIX}}^{-} \leftarrow \eta_{\text{QMIX}} \theta_{\text{QMIX}} + (1 - \eta_{\text{QMIX}}) \theta_{\text{QMIX}}^{-} \tag{25}$$

V. ILCHO MARL Framework

In this section, details of the ILCHO algorithm are introduced. The optimization problem $(\mathcal{P})$ involves K UEs competing and

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cooperating for the limited resources of the LEO satellites, which forms a general-sum game. Due to the NP-hard nature of $(\mathcal{P})$, this problem is transformed into a MARL optimization problem with the characteristics of a stochastic game. To find the optimal solution, the value-based MARL algorithm QMIX is used [32].

The objective function in $(\mathcal{P})$ is expressed as (14), aiming to enhance the NTN service performance. To solve the optimization problem $(\mathcal{P})$ through MARL, the following MDP modeling is performed.

Agent: Multiple UEs, denoted as agent k, act as agents. Each agent operates in a partially observable environment where they do not know the observations of the other agents.

State: In QMIX, the state is composed of the set of local observations from each agent k, represented as o_k^t. The local observation of agent k is represented in (26).

$$o_k^t = \left\{ I_{k,n}^t, D_{k,n}^t, L_{k,n}^t, V_{k,n}^t \mid n = 1, 2, \ldots, N_{\max} \right\} \tag{26}$$

For each UE k at time step t, $I_{k,n}^t$ is the logical index of the accessible LEO satellites, $D_{k,n}^t$ is the distance between the accessible LEO satellite, $L_{k,n}^t$ is the number of channels used by the LEO satellite, and $V_{k,n}^t$ is the remaining visible time (RVT) of the LEO satellite. $N_{\max}$ is the number of maximum visible LEO satellites for UE k. The value of $N_{\max}$ can be obtained through empirical simulation based on Lemma 2.

Action: a_k^t denotes the action of agent k at time step t. The action indicates which accessible LEO satellite is selected, with a value ranging between 1 and $N_{\max}$. For example, if $a_k = 2$, this means agent k is recommended to conduct HO to the $I_{k,2}$th LEO satellite among its accessible LEO satellites.

Reward: Since the objective function aims to maximize the NTN service performance while minimizing the average HO counts, the reward for agent k at time step t, denoted as r_k^t, is formulated as follows.

$$r_k^t = \begin{cases} -p_1, & \text{if HO success} \\ -p_2, & \text{if HOF occurs} \\ -U_{k,R}^t + U_{k,RVT}^t, & \text{otherwise} \end{cases} \tag{27}$$

In this model, $U_k(\cdot)$ represents the cost function for the reward, where $U_k(x) = c_3/(1 + e^{c_1(x-c_2)})$. The function $U_k(\cdot)$ is a sigmoid-like function, with a range of $(0, c_3)$ when c_3 is positive, and $(c_3, 0)$ when c_3 is negative. $U_{k,R}$ is defined as the throughput cost function, and $U_{k,RVT}$ is defined as the RVT cost function. For $U_{k,R}$, the parameters c_1, c_2, and c_3 are respectively set to negative, positive, and positive, so that higher throughput results in a higher reward. Conversely, for $U_{k,RVT}$, the parameters c_1, c_2, and c_3 are set to positive, positive, and negative, so that a lower RVT results in a penalty. At the first point when a LEO satellite becomes accessible to a UE, it has a high RVT but low throughput. Similarly, selecting a LEO satellite with high throughput but decreasing RVT may not be optimal. By using this sigmoid-like cost function, the model avoids the error of selecting a LEO satellite that either has high RVT but low throughput or high throughput but low RVT, balancing both factors. In addition, the reward function does not explicitly include a load balancing term, the HOF penalty indirectly encourages agents to avoid overloaded satellites, as HOFs often occur due to resource saturation. This approach aligns with the principles of reward shaping, where actions consistently leading to negative rewards (selecting a congested satellite) are naturally suppressed during training [47]. As a result, the ILCHO scheme implicitly promotes load-aware decision-making through reinforcement learning.

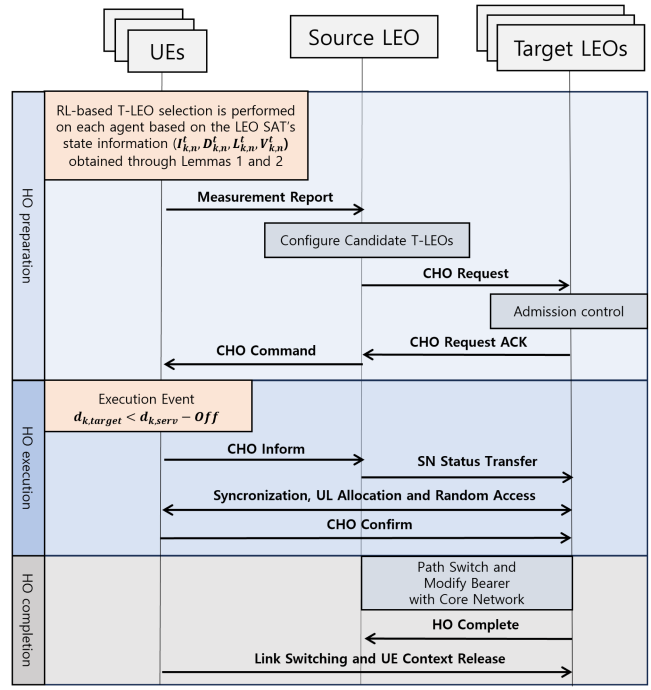

Fig. 3. Details of the ILCHO process.

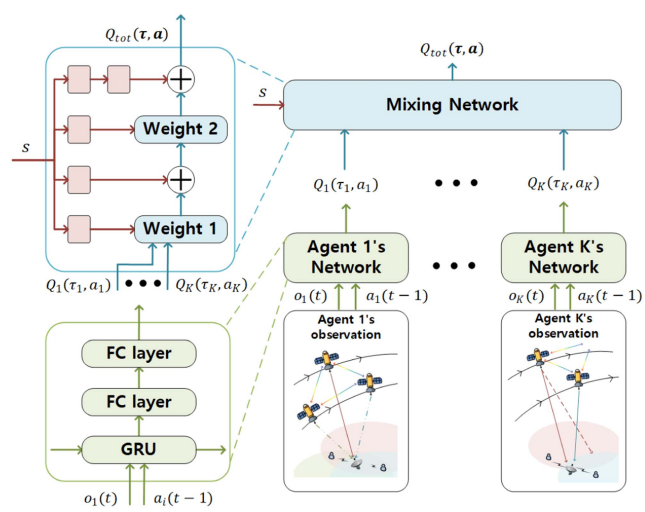

Fig. 4. ILCHO CTDE process.

As shown in Fig. 3, CHO has two events. The ILCHO scheme recommends the optimal LEO satellite as the T-LEO satellite to each UE through an RL-based action. Then, each UE performs HO when the execution event is satisfied. The execution event is set as a distance-based event applicable to the NTN and is defined as

$$d_{k,\text{tar}} < d_{k,\text{serv}} - O_{\text{off}} \tag{28}$$

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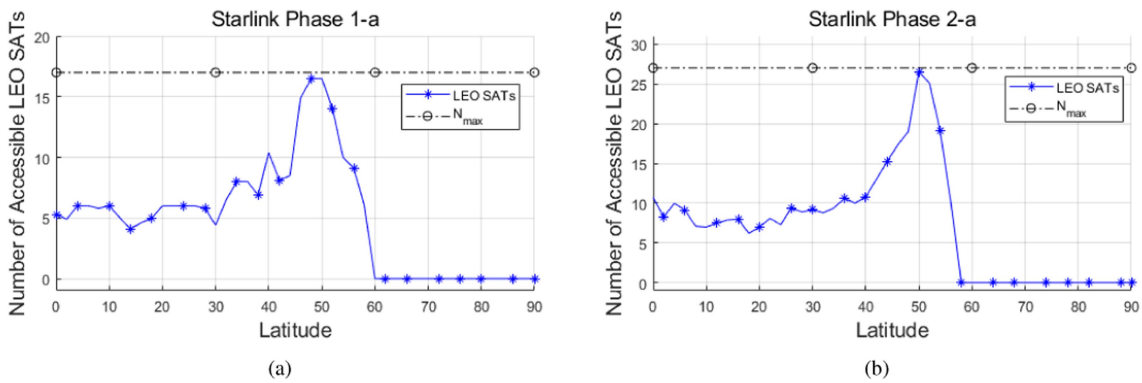

Fig. 5. Number of average accessible links based on the UE latitude, using Lemma 2 (a) in Starlink Phase 1-a and (b) in Starlink Phase 2-a.

where $d_{k,tar}$ is the distance between UE k and the T-LEO satellite recommended through the action, and $d_{k,serv}$ is the distance between UE k and the current S-LEO satellite. O_{off} is the distance offset. When condition (28) is satisfied, UE k performs HO to the T-LEO satellite. The observation o_k^t for UE k is sent to the centralized controller, and each UE selects the T-LEO satellite through action a_k^t. According to the MDP, the next observation o_k^{t+1} and reward r_k^t are obtained. For neural network training, the experience is stored in the replay buffer $\mathcal{D}$ in the form of $(\boldsymbol{o}^t, \boldsymbol{\tau}^t, \boldsymbol{a}^t, \boldsymbol{r}^t, \boldsymbol{o}^{t+1}, \boldsymbol{\tau}^{t+1})$. After randomly sampling minibatches from the replay buffer, training is performed by minimizing (23). Additionally, the target network is updated every T_t interval during training. The proposed ILCHO algorithm is summarized in Algorithm 1.

The ILCHO scheme's complexity is split into an offline training phase and a real-time CHO execution (inference) phase. According to [48], the time complexity of the training phase is

$$\mathcal{O}\left(bN_{epi}T_s\left(\frac{K(P_a+N_{\max})+P_m+P_h}{T_{update}}+\frac{P_t}{T_t}\right)\right) \quad (29)$$

where, b is the batch size, N_{epi} is the number of episodes, T_s is the service duration, T_{update} is the update interval of the main network, T_t is the update interval of the target network, K is the number of agents, and P_a, P_m, P_h, P_t are the number of parameters of the agent network, the mixnet network, and the hyper network, respectively. $N_{\max}$ is the number of actions, which is derived via Lemma 2 and can be treated as a constant. Therefore, the training time complexity in (29) can be $\mathcal{O}(K)$ when the network size and $N_{\max}$ are fixed. Likewise, the execution phase complexity is $\mathcal{O}(K(P_a+N_{\max})+P_m+P_h)$, which can be represented as $\mathcal{O}(K)$ under fixed neural network sizes and a constant $N_{\max}$.

VI. Performance Analysis

The effectiveness of the NTN performance through the ILCHO scheme is analyzed. The superiority of the proposed ILCHO algorithm is validated through experiments conducted in the Starlink Phase 1 and Starlink Phase 2 constellation environments. In Starlink Phase 2, the LEO satellites are positioned at a lower altitude (ranging between 340 to 370 km) compared to Phase 1. Due to the lower altitude, the coverage is smaller than in Phase 1, requiring the deployment of more LEO satellites and resulting in more frequent HOs. Phase 2 is an environment that requires higher algorithmic accuracy. Therefore, the initial simulation experiments focus on Starlink Phase 2, where the experimental results based on Phase 1 will be introduced in a later section. The parameters used in the simulation are shown in Table II. Assuming a rural scenario, the LoS probability values were taken from Table 6.6.2-1 in [7].

Based on Lemma 2, the average number of accessible links by latitude and the maximum value of accessible links can be observed in Fig. 5. Fig. 5(a) and (b) represent the simulation environments for Starlink Phase 1-a and Starlink Phase 2-a, respectively. The maximum number of accessible links are 17 and 27, respectively. The service region is set to $[39-41°N]$ and $[39-41°E]$ where UEs were randomly placed in each simulation. The experiment was conducted with UE numbers ranging from 5 to 100. The total service duration was set to 10 minutes with time slots spaced at 1s intervals.

In the training phase, an ϵ-greedy exploration strategy was adopted, with the ϵ value starting at a maximum of 0.9 and gradually decreasing to 0.1 as the episodes progressed. The learning rate was set to 10^{-4}, and the discount factor was 0.99. Depending on the number of UEs, the required training episodes increased from 3,000 to a maximum of 30,000. The simulation was implemented based on the Python 3.11 environment using PyTorch version 2.1.0.

Fig. 6 presents the results of applying ILCHO with centralized training and decentralized execution for 30 agents in the Starlink Phase 2-a constellation. In Fig. 6(a), the learned values for (27) are shown, where the total score (the sum of rewards obtained at each step) converges as the episode progresses. Fig. 6(b) illustrates the average number of HOs and HOFs for all agents and Fig. 6(c) displays the average throughput of all agents. As training progresses, the number of HOs and HOFs tends to decrease, and the average throughput of all agents tends to increase. The objective function $(\mathcal{P})$ aims to maximize the overall service performance while reducing the number of HOs, confirming that the learning-based ILCHO scheme operates effectively.

To validate the effectiveness of the proposed sigmoid-based reward function, an additional experiment with a linear-based

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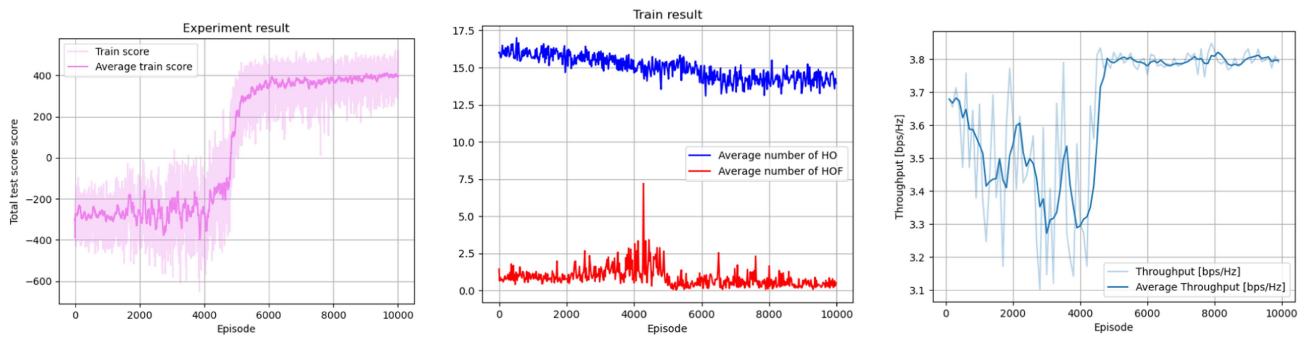

(a) (b) (c)

Fig. 6. Training results per episode with 30 agents in the Starlink Phase 2-a constellation environment when using a **sigmoid-based reward function.** (a) Total train score, (b) average number of HO per agent, and (c) average throughput per agent.

TABLE II
PARAMETERS OF THE SIMULATION

Simulation Parameters	
Region of interest	[39-41°N, 39-41°E]
Number of UEs	5 - 100
Deployment scenario	Rural
LoS probability, Shadow fading σ_{S_F},	Table 6.6.2-1
and Cluster loss C_L	in [6]
UE antenna configuration	VSAT
Frequency band	Ka-band
Minimum elevation angle ψ_{min}	30°
Channel bandwidth	250 MHz
E_{EIRP}	-4 dBW/MHz
K_B	-228.6 dBW/K/Hz
G_T	15.9 dB/K
Service duration T_s	10 min
Maximum number of channels J	8 [49]
Constellation Parameters (Phase 1, Phase 2)	
Altitude of LEO satellite H_L	550 km, 340 km
Inclination angle α	53°, 53°
Number of orbital planes N_p	72, 48
Satellite per plane N_L	22, 110
Total number of satellite	1584, 5284
Hyper-Parameters of ILCHO	
Learning rate β	10^{-4}
Discount factor σ	0.99
Batch size b	32
Epsilon ϵ	0.9 - 0.1
Numbers of training episodes N_{epi}	3,000 - 30,000
Target update interval T_t	10
Soft update coefficient η_{QMIX}	0.001

reward function was conducted. In Fig. 7, the performance of training with a linear reward function is analyzed under the same experimental conditions as Fig. 6 (i.e., 30 agents in the Starlink Phase 2-a constellation environment). For this comparative analysis, the sigmoid-based reward function from equation (27) was replaced with the following linear reward function

$$r_k^t = \begin{cases} -p_1, & \text{if HO success} \\ -p_2, & \text{if HOF occurs} \\ w_1 R_{NTN} + w_2 T_{RVT}, & \text{otherwise} \end{cases} \tag{30}$$

where T_{RVT} denotes the RVT of the serving LEO satellite, and w_1 and w_2 are the weighting factors for throughput and RVT, respectively. The training results using the linear reward function are presented in Fig. 7. While the total reward in Fig. 7(a) shows a trend of convergence as the episodes progress, the number of HOF in Fig. 7(b) and the throughput in Fig. 7(c) fail to converge, exhibiting significant fluctuations and remaining unstable.

This performance gap arises from the inherent orbital characteristics of LEO satellites non-linear trade-off between throughput and RVT. After becoming accessible to a user, a satellite approaches until it reaches its point of minimum distance, after which it moves away. Consequently, throughput is a non-monotonic function of time, increasing and then decreasing. In contrast, RVT is a monotonically decreasing function from the moment the satellite becomes accessible.

A linear function struggles to find the optimal point for this complex relationship. It cannot effectively balance a decreasing RVT with a throughput that first increases and then decreases. For instance, it might undervalue a satellite simply because its RVT is decreasing, even before the throughput reaches its peak. Conversely, it could overvalue a satellite that has high throughput but is about to leave the service area (low RVT).

Conversely, our sigmoid-based cost function is specifically designed to navigate this trade-off. It creates a "sweet spot" by rewarding states where both throughput and RVT are favorable, while penalizing states at the extremes (e.g., very low throughput despite high RVT, or high throughput with critically low RVT). This encourages the MARL agent to learn a more robust and farsighted handover policy that optimizes to sustain the service quality rather than focusing on a single, linearly weighted metric.

In order to evaluate the performance of the ILCHO scheme, a comparison with the two HO strategies of [22] and [26] are conducted. In a TN, it is easy to compare the measurement values between cells based on the RSRP. However, in an NTN, the RSRP values between cells are often very similar and unpredictable since the LEO satellites are at a remote

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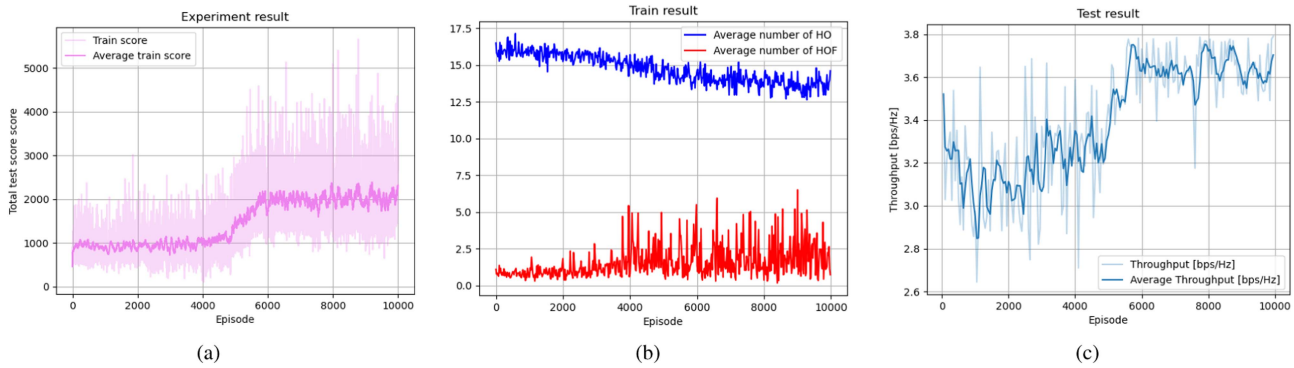

Fig. 7. Training results per episode with 30 agents in the Starlink Phase 2-a constellation environment when using a **linear reward function.** (a) Total train score, (b) average number of HO per agent, and (c) average throughput per agent.

distance. According to 3GPP Release 16 TR 38.821 [7], using the measurement-based HO event A3 (which is widely used on the ground for UE HO in TNs) may cause difficulties for UEs in finding the optimal cell in NTNs. Event A3 HO is triggered when the signal power received from the neighboring base station exceeds the signal power received from the serving base station by an offset margin. Therefore, two benchmark HO strategies are triggered by the distance-based event A2 (which requires the serving LEO satellite's signal level to become smaller than the threshold)

$$d_{\text{serving}} + O_{\text{off}} < \zeta \tag{31}$$

where ζ is the distance threshold, which is set as the distance at which the serving LEO satellite will no longer be accessible if d_{serv} exceeds ζ. The two benchmark HO strategies are as follows.

- *Minimum distance (MD) based HO:* The UE attempts HO to the LEO satellite located at the shortest distance among the accessible LEO satellites.
- *Maximum visible time (MVT) based HO:* The UE attempts HO with the LEO satellite with the longest RVT.

In addition, representative HO schemes from two key papers are described in the following and compared in the performance analysis.

- *Load balancing satellite HO (LBSH) [26]:* The LBSH strategy uses multi-agent Q-learning, which performs learning to minimize the number of HOs. The reward is set as a combination of the RVT and lower load, and is designed to receive a large penalty when HO is performed.
- *HO strategy based on network flows (HSNF) [22]:* A network flow model based on graph theory is proposed, which performs HO based on a cost matrix and considering the flow between each UE and LEO satellite based on the average throughput and remaining bandwidth.

Fig. 8 presents a comparison of the 4 HO strategies mentioned above. In this paper, HOF occurrence is defined due to the following reasons: 1) attempting HO to a target LEO satellite that lacks sufficient resources, 2) attempting HO to a target LEO satellite that no longer has RVT, or 3) attempting HO to a LEO satellite that is not accessible. The goal of this research is to find the optimal cell among numerous LEO satellites.

Fig. 8(a) shows the average number of HOs according to the number of UEs during the service duration. The average number of HOs of all strategies increases with the number of UEs. This tendency is due to the limited resources of each LEO satellite, which restricts the number of UEs that can be served by a single cell. Consequently, as the number of UEs increases, the complexity of selecting a target cell rises, leading to an increase in the average number of HOs. Despite this, ILCHO and HSNF show only a slight increase. For ILCHO, with 5 agents, the average HO count is 10.8 per service duration (HO conducted approximately every 55.56 s), and with 100 agents, it is 12.95 per service duration (HO conducted approximately every 46.32 s). LBSH simulation was conducted in scenarios with a very small number of satellites. However, when applied to the Starlink constellation, the problem of an exponentially increasing action space occurs. Although the learning scores converged, the MD, MVT, and LBSH schemes do not achieve an optimal performance, which results in a large number of HOs.

Fig. 8(b) shows the average number of HOFs per service duration based on the number of UEs. For the MD-based HO and MVT-based HO, these strategies perform well with fewer UEs but exhibit an increase in HOFs as the number of UEs increases. This tendency can be attributed to the relatively short distances between UEs compared to the distance to the LEO satellites, which leads to a high correlation between the HO criteria, such as distance and visible time. Consequently, when considering only the distance and RVT in making HO decisions, the results show that this would be inadequate. Similarly, LBSH also demonstrates a high number of HOFs for the same reasons mentioned. HSNF was also applied to a constellation with a very small number of satellites. When the HSNF algorithm is applied to commercial mega-constellations, calculating the link cost between every UE and every LEO satellite in real-time presents a significant challenge, which may not yield optimal results. Consequently, some unnecessary HOFs are observed. In contrast, ILCHO demonstrates a stable performance even as the number of agents increases. Specifically, with up to 50 agents, ILCHO experiences 0.48 HOFs per service duration, and with 100 agents, it results in 1.38 HOFs per service duration. This indicates that the ILCHO scheme can maintain a reliable performance despite an increase in supporting UEs.

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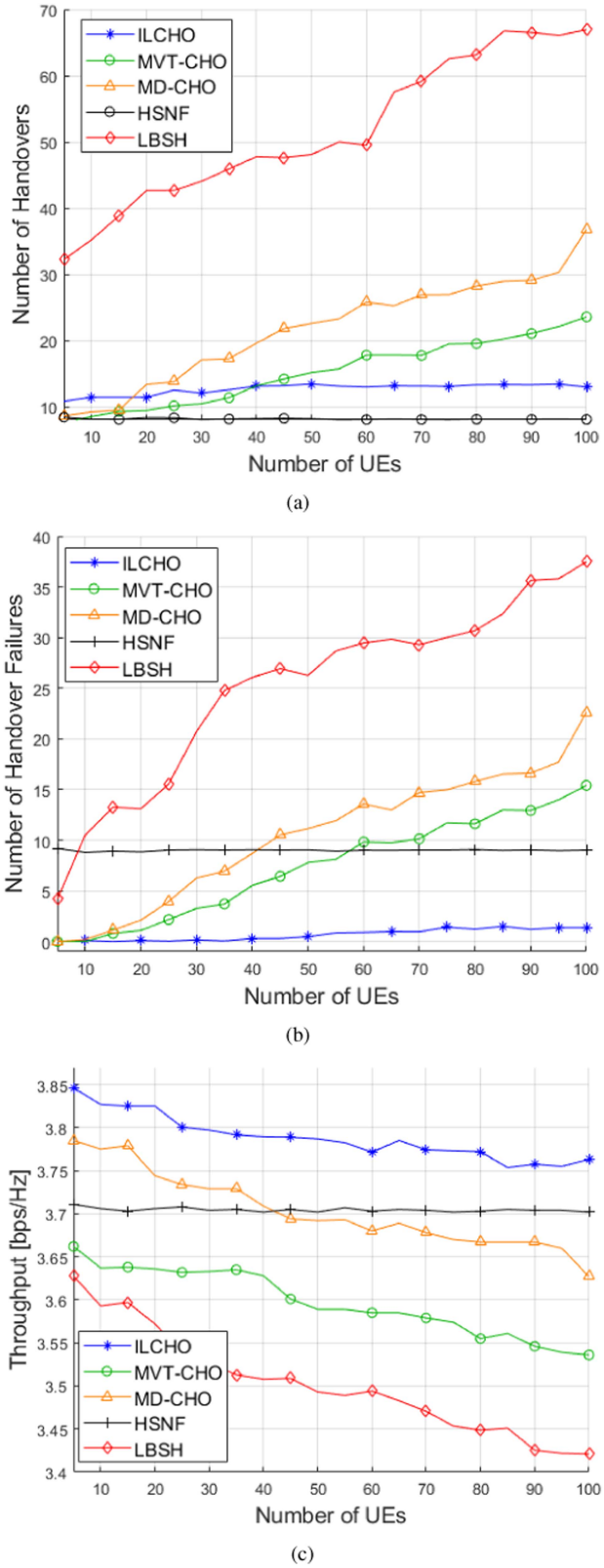

Fig. 8. Simulation results in the Starlink Phase 2-a constellation environment, (a) average number of HOs per agent, (b) average number of HO failures per agent, and (c) average throughput per agent.

Fig. 8(c) shows the average throughput of all agents over the service duration. All strategies tend to experience a decrease in throughput as the number of UEs increases. As the ILCHO scheme aims to maximize the NTN service performance, it achieves throughput values of 3.85 bps/Hz with 5 agents, 3.79 bps/Hz with 50 agents, and 3.76 bps/Hz with 100 agents, which is a superior performance compared to the other algorithms.

Additional experiments were conducted to confirm the effectiveness of the ILCHO scheme based on Starlink Phase 1, which has a relatively higher altitude compared to Phase 2 satellites. The simulation results presented in Fig. 9 show similar trends to those observed in Phase 2. When applying Phase 1, the main difference compared to Phase 2 is that the altitude of LEO satellites increases from 340 km to 550 km, increasing the visible time of the LEO satellites. Additionally, the total number of LEO satellites decreases significantly, leading to a reduction in the number of accessible LEO satellites, as observed in Fig. 5. Fig. 9(a), (b), and (c) represent the average number of HOs, the average number of HOFs, and the average throughput of all agents, respectively.

In Fig. 9(a), both ILCHO and HSNF algorithms show stable HO counts regardless of the number of UEs. ILCHO demonstrates a range from a minimum of 8 HOs per service duration (HO conducted approximately every 75 s) to a maximum of 9.17 HOs per service duration (HO conducted approximately every 65.43 s). Phase 1, with its higher LEO altitudes (compared to Phase 2), results in a longer visible time for the LEO satellites, which reduces the HO count. In contrast, MVT algorithms, while showing fewer HO and HOFs counts with fewer agents, exhibit minimal differences starting from 15 agents compared to Phase 2. This is attributed to the increased visible time but reduced number of visible LEOs, leading to a decrease in algorithm stability. Specifically, the stability of the MD scheme significantly drops due to the reduced number of LEO satellites due to scenarios in which UE correlation is high. In Fig. 9(b), the ILCHO scheme demonstrates a superior performance with fewer HOF counts than 0.98 per service duration. While the NFB scheme shows a reduction in HOFs, the other algorithms experience an increase in HOFs compared to Phase 2. This indicates that the reduction in the average number of accessible LEO satellites significantly impacts the HOFs.

Fig. 9(c) demonstrates that the ILCHO scheme can achieve a higher throughput compared to all other schemes. The results indicate that the ILCHO scheme can perform well even within current LEO satellite commercial mega-constellations. Particularly, by leveraging the TLE data (e.g., distance and elevation angle) rather than relying on measurement-based triggering, the ILCHO scheme demonstrates its ability to achieve a superior performance.

Fig. 10 presents the experimental results of the ILCHO scheme within the Starlink Phase 1-a, Phase 2-a, and hybrid constellation where Phase 1-a and Phase 2-a coexist. The results for the number of HOs (Fig. 10(a)), number of HOFs (Fig. 10(b)), and average throughput (Fig. 10(c)) are shown

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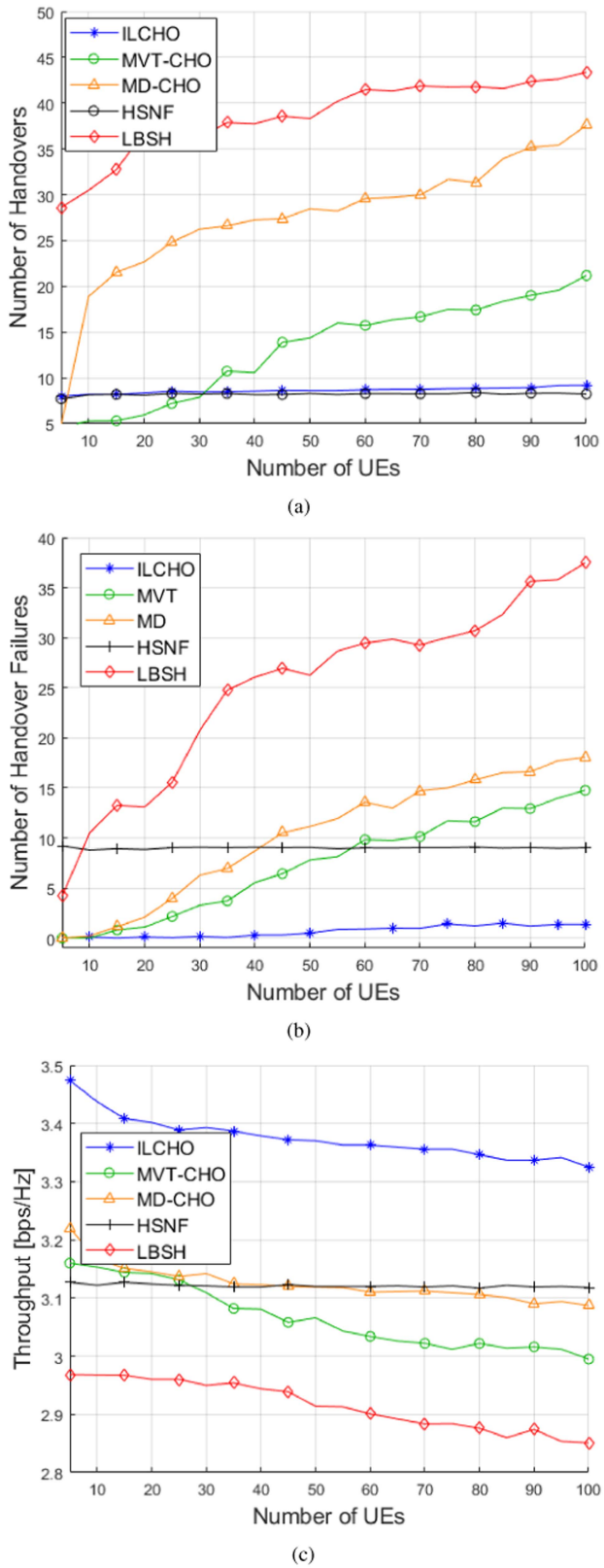

Fig. 9. Simulation results in the Starlink Phase 1-a constellation environment, (a) average number of HOs per agent, (b) average number of HO failures per agent, and (c) average throughput per agent.

TABLE III
NUMBER OF HOS, HOFS, AND AVERAGE THROUGHPUT IN THE HYBRID CONSTELLATION

Number of HOs (Agent 5 to 100)			
	Average	Minimum	Maximum
Phase 1-a	8.64	8.00	9.20
Phase 2-a	12.67	10.80	13.42
Hybrid	12.24	9.00	13.16
Number of HOFs (Agent 5 to 100)			
	Average	Minimum	Maximum
Phase 1-a	0.53	0.00	0.97
Phase 2-a	0.67	0.00	1.50
Hybrid	0.97	0.53	1.47
Average Throughput (Agent 5 to 100)			
	Average	Minimum	Maximum
Phase 1-a	3.375	3.325	3.474
Phase 2-a	3.788	3.754	3.846
Hybrid	3.647	3.572	3.748

for each constellation. The average, minimum, and maximum values for each of these metrics are summarized in Table III.

The hybrid constellation simultaneously reflects the characteristics of both Starlink Phase 1-a and Starlink Phase 2-a. It inherits the low HO count characteristic of Starlink Phase 1-a, demonstrating a lower number of HOs compared to Phase 2-a. Furthermore, it leverages the high-throughput advantage of Phase 2-a, achieving higher throughput than Phase 1-a. In the case of HOFs, it can be observed that there is no significant difference among the three constellations.

A. Fairness and Load Balance

This part addresses the fairness and load balance issues of the ILCHO scheme. The fairness of the throughput of each UE is analyzed through Jain's fairness index (JFI), which is defined as

$$J_{JFI} = \frac{(\sum_{k=1}^{K} R_k)^2}{K \sum_{k=1}^{K} (R_k)^2} \tag{32}$$

where, $R_k = \sum_{t \in T} \sum_{s_m \in \mathcal{S}} R_{m,k}$ denotes the average throughput of user u_k during the service duration. A higher JFI indicates fairer resource allocation.

Fig. 11 shows the cumulative distribution function (CDF) of the throughput of 100 agents in Starlink Phase 2-a. The JFI is 0.9798, which is close to the perfect fairness of 1.0. This quantitatively proves that our proposed MARL approach leads to fair resource allocation. The CDF of the throughput in Fig. 11 shows that starvation of edge UEs does not occur.

Channel occupancy for LEO satellites s_m at time t is defined as $O_{m,t} = L_{m,t}/J$, where $L_{m,t} = \sum_{k=1}^{K} l_{m,k}$ is the number of connected UEs. The variance of channel occupancy is used as the load balance metric. This variance is calculated across the

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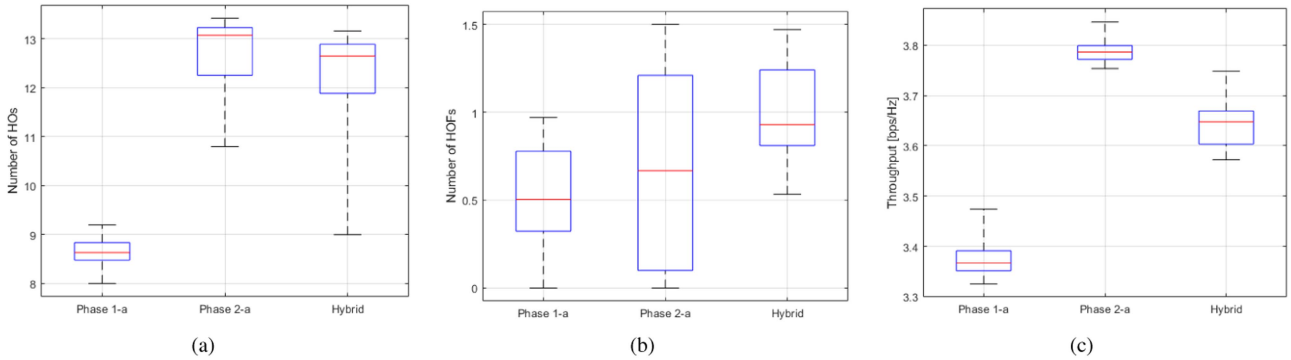

Fig. 10. Simulation comparision results of the ILCHO scheme for Starlink Phase 1-a, Phase 2-a, and hybrid constellation with the number of agents ranging from 5 to 100. The red line indicates the median value, the box represents the interquartile range (25th to 75th percentile) in the performance graphs of (a) average number of HOs per agent, (b) average number of HO failures per agent, and (c) average throughput per agent.

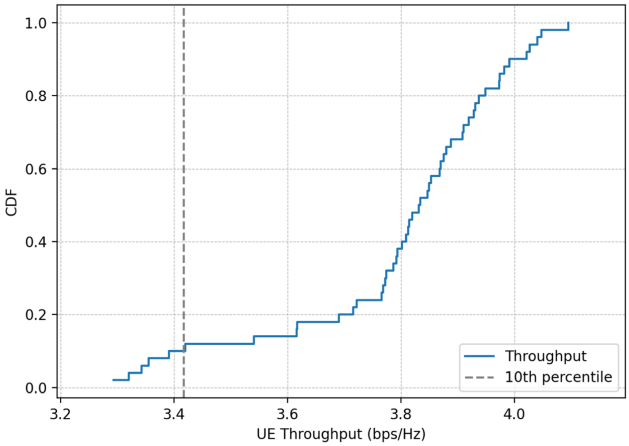

Fig. 11. Throughput CDF of 100 agents in Starlink Phase 2-a. $J_{JFI} = 0.9798$, 10th-percentile throughput = 3.50 8bps/Hz, median throguhput = 3.846bps/Hz, average throughput = 3.757 bps/Hz.

TABLE IV
VARIANCE OF CHANNEL OCCUPANCY OF LEO SATELLITES WITH 50 AGENTS IN STARLINK PHASE 2-A

	ILCHO	MD-CHO	MVT-CHO	LBSH	HSNF
Variance	0.072	0.372	0.400	0.077	0.089

common pool of accessible LEO satellites to the UE population in the simulation region. Table IV shows the variance of channel occupancy of LEO satellites with 50 agents in Starlink Phase 2-a.

This superior load balancing is a direct result of the agents learning through the reward function. Concentrating UEs on a single highly-loaded satellite increases the probability of HOF, and this event incurs a significant penalty. Thus, the agents inherently learn to distribute the load to maintain service stability.

VII. CONCLUSION AND FUTURE WORK

In this paper, the ILCHO scheme is proposed to support HO decisions in an NTN of LEO satellites. As LEO satellites move at very high speeds and have long latency (due to their orbiting altitude) compared to TNs, it is difficult to apply traditional TN HO algorithms that use measurement-based triggering events. The proposed ILCHO scheme maintains accurate control through the derived state information of the LEO satellites (in regards to position and orbital path) which is used to verify the conditions for a LEO satellite to be accessible to enable optimized target LEO satellite selection that will maximize the overall network performance while minimizing the number of HOs to reduce network overhead. To enable consistent optimized HOs to multiple UEs simultaneously, the ILCHO scheme uses LEO satellite movement predictions made by its MARL system. The ILCHO scheme makes optimized satellite selections during the HO preparation event where HO is performed through a distance-based execution event. As a result, a more stable number of HOs and higher throughput is obtained compared to other NTN HO schemes. Future research needs to be conducted while considering other performance degrading variants in LEO satellite NTNs, where the control problem will become more complex and scalable optimization control may become more challenging.

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